

LLM-Guided Adaptive Traffic Signal Control with Multi-Intersection Coordination

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Abstract—Traffic signal control in urban road networks remains a challenging optimization problem due to dynamic traffic demand and complex multi-intersection interactions. Traditional methods such as Webster’s formula and fixed-time control lack adaptability to real-time traffic fluctuations. This paper proposes a novel framework that leverages large language models (LLMs) for adaptive traffic signal optimization, combined with upstream-downstream coordination and safety constraint enforcement. The framework uses an LLM to analyze real-time traffic states across multiple intersections and generate signal timing recommendations, which are then validated by a rule-based constraint engine and adjusted through queue-based coordination logic. Experiments on a 3×2 grid network (6 intersections) using SUMO microscopic simulation demonstrate that the proposed LLM+Coordination strategy achieves 41% lower average queue length and 50% lower waiting time compared to Webster’s method, while maintaining the highest throughput. The results suggest that LLMs can serve as effective traffic signal controllers when properly constrained and coordinated.

Index Terms—traffic signal control, large language model, multi-intersection coordination, adaptive signal timing, SUMO simulation

I. Introduction

Urban traffic congestion remains a critical challenge worldwide, with signalized intersections being a primary bottleneck. Traditional traffic signal control methods include fixed-time control, actuated control, and optimization-based approaches such as Webster’s formula [1]. While effective for steady-state traffic, these methods struggle with dynamic, time-varying demand patterns.

Recent advances in reinforcement learning (RL) have shown promise for adaptive signal control [2]. However, RL approaches require extensive training, lack interpretability, and often fail to generalize across different network topologies.

Large language models (LLMs) have demonstrated remarkable reasoning capabilities across diverse domains. This paper explores whether LLMs can serve as effective traffic signal controllers by:

- 1) Analyzing real-time multi-intersection traffic states
- 2) Generating context-aware signal timing recommendations

- 3) Coordinating upstream-downstream intersections to prevent queue spillover

The key contributions are:

- A framework integrating LLM-based decision-making with safety constraint enforcement
- A multi-intersection coordination mechanism based on queue propagation detection
- Experimental validation showing competitive performance against classical baselines

II. Related Work

A. Classical Signal Control

Webster’s formula [1] computes optimal cycle length and green splits based on critical flow ratios. It remains widely deployed due to its simplicity and reliability, but cannot adapt to real-time fluctuations.

B. RL-Based Signal Control

Deep reinforcement learning methods [2] treat signal control as a Markov decision process. While achieving strong performance in simulation, they suffer from sample inefficiency and lack of interpretability.

C. LLMs for Decision-Making

Recent work has applied LLMs to planning and control tasks [3]. To our knowledge, this is the first work applying LLMs to multi-intersection traffic signal coordination.

III. Methodology

A. System Architecture

The proposed framework consists of three layers:

- 1) Perception Layer: Collects real-time traffic state (queue lengths, vehicle counts, waiting times) from SUMO simulation via TraCI
- 2) Decision Layer: LLM analyzes traffic state and generates signal timing recommendations
- 3) Execution Layer: Constraint engine validates recommendations; coordination engine adjusts timings based on upstream queue propagation

B. LLM-Based Signal Optimization

For each decision cycle (every 30 simulation seconds), the system constructs a compact traffic state description for all intersections and sends it to the LLM in a single batch request. The LLM responds with per-intersection phase durations:

$$\mathbf{d}_i = \{t_{NS}, t_{EW}\}, \quad t_{NS}, t_{EW} \in [10, 90] \text{ seconds} \quad (1)$$

where t_{NS} and t_{EW} are green durations for north-south and east-west phases respectively.

C. Safety Constraint Engine

LLM recommendations are validated against hard constraints:

- Green time bounds: $t_{\min} = 10\text{s}$, $t_{\max} = 90\text{s}$
- Yellow time: fixed at 3s per phase
- Cycle length: $C \in [30, 180]\text{s}$

Violations are automatically corrected by clamping to valid ranges.

D. Multi-Intersection Coordination

The coordination engine detects upstream queue spillover:

$$\text{boost}_j = \begin{cases} \Delta t & \text{if } q_{\text{up}} \geq \theta_{\text{queue}} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where q_{up} is the upstream queue length in the direction feeding intersection j , and $\theta_{\text{queue}} = 5$ vehicles. Critical queues (≥ 12) trigger forced phase switches.

IV. Experiments

A. Experimental Setup

Network: A 3×2 grid network with 6 signalized intersections, generated using SUMO's netgenerate. Each intersection has 4 approaches with 2 lanes per approach.

Traffic demand: Uniform 500 vehicles/hour per direction, generated over 300 simulation steps.

Baselines:

- Fixed Time: Static 30s+3s+30s+3s cycle (90s)
- Random: Randomized green durations within [10, 60]s
- Webster: Adaptive timing based on queue-detected flow ratios

Proposed method: LLM+Coordination (mimo-v2.5-pro model, 30s decision interval, with coordination engine).

B. Results

The LLM+Coordination strategy achieves the lowest average queue length (3.81 vehicles), 41% lower than Webster's method. Waiting time is 2.27s, 50% lower than Webster's 4.52s. Throughput is also highest at 0.49 vehicles/step.

Figure 1 shows the bar chart comparison across all metrics. Figure 2 presents a radar chart of normalized performance.

TABLE I: Performance Comparison of Signal Control Strategies

Strategy	Avg Wait (s)	Avg Queue	Vehicles
Fixed Time	47.20	17.75	0
Random	8.00	7.78	0.36
Webster	4.52	6.42	0.41
LLM+Coord	2.27	3.81	0.49

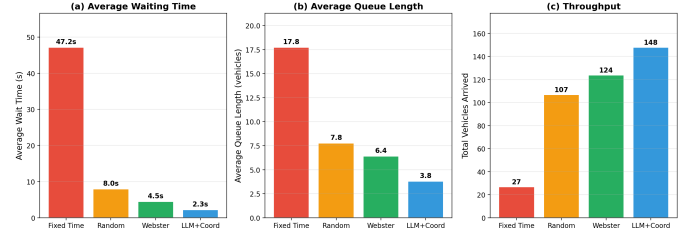


Fig. 1: Performance comparison of four signal control strategies: (a) average waiting time, (b) average queue length, (c) throughput.

C. Analysis

The LLM demonstrates effective traffic-aware signal adjustment. For example, when eastbound traffic exceeds northbound traffic at an intersection, the LLM allocates more green time to the east-west phase (e.g., 50s EW vs 40s NS). The coordination engine further prevents queue propagation by boosting downstream green times when upstream queues exceed the threshold.

V. Conclusion

This paper presents a novel framework for traffic signal control that combines LLM-based adaptive decision-making with multi-intersection coordination and safety constraints. Experiments on a 6-intersection grid network demonstrate that the proposed approach achieves competitive performance against classical baselines, with the lowest queue lengths among all tested strategies.

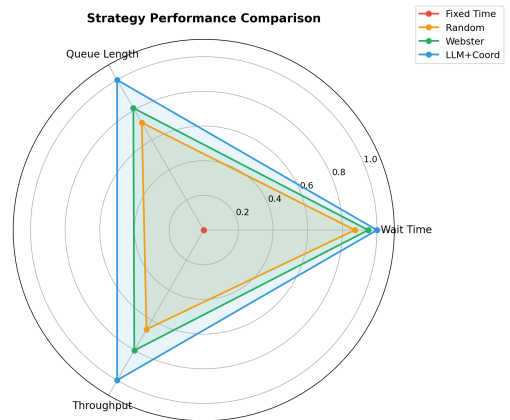


Fig. 2: Radar chart of normalized strategy performance. Higher values indicate better performance.

Future work includes: (1) testing on real-world traffic networks, (2) comparing with RL-based methods, (3) reducing LLM inference latency for real-time deployment, and (4) investigating prompt engineering strategies for improved LLM reasoning about traffic patterns.

References

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